Apertus Hybrid

Technical report · Hack Apertus · Track 2A (OST) · booklet-grounded multilingual NLI · team Arroztek · code Apache-2.0, report CC-BY-4.0

1 Summary

Given a Swiss voting booklet (PDF) and a claim in German, French or Italian, our system returns the official label — **0 entailment, 1 neutral ("not established"), 2 contradiction** — with calibrated class probabilities, the exact source passages the model read (page, offsets, PDF SHA-256), the **actual token usage** and the **actual inference time**. Document and claim languages may differ.

Our proposal is a **frozen-head decision pipeline**: Apertus v1.5 8B is used unchanged (**no LoRA, no fine-tuning**). We read its last-layer, last-token hidden state and classify it with a small, training-only **calibrated linear head**, fed by a **hybrid (BM25 + multilingual E5) retrieval context** packed into an 8,192-token budget. We compare it against "plain Apertus", i.e. the same frozen model asked to pick an answer from the same context.

0.949	**0.942**	**8,060**	**2.52 s**
Macro-F1, hybrid-8k head (plain Apertus, same context: 0.549)	cross-language Macro-F1 (n = 190 of 276)	mean prompt tokens, −42.9 % vs the full/capped 14,106	mean cached inference time per example (A40/A6000-class GPU; see §4 for what it includes)

> **Evidence level.** All headline numbers are **development validation** (276 rows, 207 distinct claim/document pairs, 3 voting events). hybrid-8k was chosen on this same validation set, so there is **no fresh held-out score** for the final head; they are not organizer-hidden results. The historical 310-row internal holdout (0.8999) belongs only to the earlier full-context head.

2 Architecture: frozen heads + hybrid decision pipeline



Why frozen heads

The base model's answer-token scores (what an API or chat endpoint exposes) are a poor NLI decision rule on this task: with the same hybrid-8k context, the restricted option-score reaches 0.549 Macro-F1. The information is in the model's internal state: a linear head on the 4,096-d last-token hidden state reaches 0.949 on the same inputs and the same 276 examples, without touching a single Apertus weight. The head is cheap (no extra Apertus forward pass), auditable (hash-pinned parameters in `track_2a/deployment/head-v15.json`) and reversible.

Hybrid retrieval and context packing

Full booklets (≈14k tokens even after a 48,000-byte cap, which truncated all 276 validation inputs) are replaced by an 8k selection. BM25 captures exact numbers and terms; multilingual E5-small (`intfloat/multilingual-e5-small`, retrieval only, never the decision) captures cross-lingual meaning. Rankings are fused by reciprocal rank; *whole* source passages are packed using the official tokenizer, including system/claim/assistant prefix, so every quote shown to the user is an exact substring of the extracted page text. No reference string or label enters inference.

Head training and calibration

Standardized multinomial logistic regression, fitted on 902 training examples with the **context matched to the one used at inference** (hybrid-8k). Regularization C and a temperature are selected with training-only connected-group out-of-fold evaluation (selected C = 0.1, temperature 1.582); validation and test labels are never used for fitting. Training has only two connected groups, which limits confidence in calibration and generalization (stated in §5).

Why a generic Apertus API key is not enough

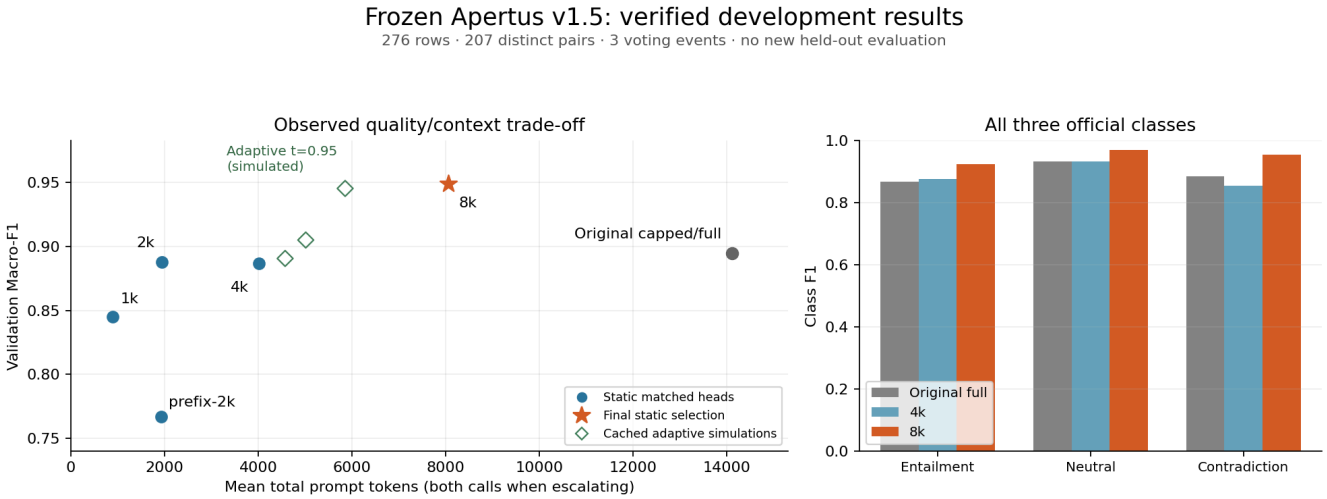
The pipeline needs the model's hidden states and our packed context; a generic base-model chat/completion endpoint returns text or token scores only, so the head cannot attach and the reported classifier is not reproduced. We therefore ship a native backend (`track_2a/scripts/serve_v15.py`) that loads Apertus v1.5 and the head, and the app refuses to invent a label when it is not connected.

3 Results (development validation, 276 examples)

Plain Apertus vs. frozen head, same frozen model, matched context

Context	Plain Apertus (option score)	Frozen head (ours)	Gain	Mean prompt tokens	Cross-language F1 (head)
hybrid-1k	0.4990	0.8447	+0.346	892.8	0.7955
hybrid-2k	0.5267	0.8876	+0.361	1,941.0	0.8428
hybrid-4k	0.5740	0.8869	+0.313	4,017.4	0.8536
hybrid-8k (selected)	**0.5488**	**0.9491**	**+0.400**	**8,059.6**	**0.9418**
original full/capped	0.6716	0.8945	+0.223	14,106.1	0.8738

"Plain Apertus" = frozen v1.5 restricted option score (softmax over the answer options A/B/C, label mapping fitted on training data). Head = standardized logistic head on the last-token hidden state. All rows: same 276 validation examples. Source: `experiments/apertus-v15-phase2/`.



Left: Macro-F1 vs mean prompt tokens (adaptive points are cached simulations, not a deployed router). Right: per-class F1 for original-full, 4k and 8k heads.

Per class and error structure (hybrid-8k head)

Class	Precision	Recall	F1	Support	Predicted 0 / 1 / 2
0 Entailment	0.867	0.988	0.924	86	85 / 1 / 0
1 Neutral ("not established")	0.989	0.949	0.969	98	5 / 93 / 0
2 Contradiction	1.000	0.913	0.955	92	8 / 0 / 84

The remaining 14 errors are almost all "over-supporting": the head says entailment where the label is neutral or contradiction (13 of 14). Language slices (claim language) and the nine document→claim pairs, deduplicated 207-pair F1 (0.9456) and cross-language results are in `final-results.json`.

Confidence is informative (calibrated head)

Validation ten-bin ECE 0.0284, Brier 0.0705, NLL 0.1393 (original full head: ECE 0.0828). Of the 210 verdicts with confidence ≥ 0.95, 208 are correct. The head's temperature was fitted on training folds only. Class probabilities come from a calibrated posterior, with no guarantee on new booklets.

4 Token usage and inference time

Every response carries the **server-measured** input-token count and inference time; none is estimated. Our context policy cuts prompt tokens by **42.9 %** (8,059.6 vs 14,106.1 mean) at higher accuracy. Time numbers below are reported separately by measurement basis; we do **not** claim the 8k system is faster or cheaper than the original full-context system.

Configuration (validation, mean per example)	Prompt tokens	Time	What the time includes
hybrid-1k head	892.8	0.523 s	cached GPU encoding + measured retrieval + CPU head (matched-head context study); not full HTTP/PDF latency
hybrid-2k head	1,941.0	0.777 s	
hybrid-4k head	4,017.4	1.365 s	
hybrid-8k head (selected)	**8,059.6**	**2.520 s**	
prefix-2k (document-order control)	1,920.9	0.453 s	
original full/capped, fresh A6000 recheck	14,106.1	3.316 s	actual integrated prediction, same GPU family (the 3.635 s A40 figure used cached encoding + CPU head)
original head on 310-example internal holdout	14,070.0	3.281 s	actual inference + measured retrieval, RTX A6000

Live application (recorded real session, 72-page official booklet)

Claim (language)	Result shown	Input tokens	Reported time	Passages
DE: *Die schweizerische Währung ist der Franken.*	Supported 97.5 %	8,072	2,271 ms	25
FR: *Le contre-projet direct propose d'inscrire … le franc.*	Supported 97.3 %	8,149	2,253 ms	29
IT: *Il controprogetto diretto stabilisce che … è l'euro.* (test claim)	Contradicted 70.3 %	8,161	2,256 ms	26

App-reported inference time includes retrieval. Start-up, weight download/verification and PDF extraction are excluded. Cost: all Phase-2 GPU experiments cost USD 1.51 of rented compute (Vast.ai; A40 and RTX A6000, 48 GB); the head adds no extra Apertus forward pass. These three claims are demonstration examples, not an accuracy test.

5 Grounding controls, limitations and honesty

Does the verdict depend on the document?

Condition (final head, validation)	Original-label Macro-F1
Correct booklet	0.949
Wrong-event booklet (seed 42 / seed 1337)	0.468 / 0.453
Empty booklet	0.273
Claim-only head (separately trained)	0.700 (earlier full-context study)

Replacing or removing the booklet cuts the score sharply, so the decision depends on the document, but with limits: the claim-only head already reaches 0.700, so claim wording carries label signal, and scores on wrong/empty booklets use the original labels as retention targets, not true labels for the intervened premise. Our registered gate result is **"document dependence, with limitations"** (GO_WITH_LIMITATIONS).

What we do not claim

- No organizer-hidden or independent held-out evaluation of the final head; hybrid-8k was selected on the same 276-row validation set (an exploratory accuracy extension of the registered 4k efficiency choice).
- Few independent events: 276 rows are 207 distinct pairs from 3 voting events; train has two connected groups. No population confidence interval is asserted, and calibration/generalization confidence is limited.
- Cited passages are the **source context the model read**, not validated minimal proof or causal attribution. In the cheap evidence-only audit, 9 of 12 strictly reduced contexts preserved the decision at 8k (3 of 12 at 4k; different cohorts, descriptive only).
- Booklet-wide "not established" remains hard to prove with retrieved context; neutral reference strings can be unrelated to the booklet. No OCR (scanned PDFs need text). Fixed 8k context can omit later evidence.

Native v1.5 needs the pinned Swiss Transformers fork and a GPU; CPU runs (smoke checks only) are slow and not benchmarked.

Where the gain comes from

Against the plain restricted base score on the same context, the head fixes 103 validation decisions and breaks 4. Most of the gain is on neutral: plain Apertus gets 13 of 98 neutral claims right (it labels most as contradiction); the head gets 93 of 98. We report this as decision-mapping evidence — it shows the information is in the frozen representation — not as proof that the weights cannot do numeric/semantic equivalence or that LoRA would not help.

6 Entailment CLI, cited passages and reproducibility

```
python -m ost_nli predict BOOKLET.pdf "CLAIM"        # single claim, JSON out
docker compose up --build -d                          # web UI on http://localhost:8000
```

Output includes `label`, `label_name`, class probabilities, input tokens, inference time and `evidence`: for each packed passage the one-based page, exact extracted text, paragraph ID, source PDF SHA-256 and page-text character offsets. A batch mode takes JSONL with `id` / `document` / `claim`. The web UI shows the same fields, flags truncated context, and downloads the JSON. Three real GPU PDF-CLI cases cover DE→FR, FR→IT and IT→DE claim/document pairs; a real browser session with six forward passes reproduced labels 0/0/2.

Reproduce

- **Application (no GPU):** hash-locked Docker image; PDF extraction, passages and retrieval work; without a backend the app refuses to classify rather than invent output. 41 core tests pass on host and in Docker.
- **Classifier backend:** Python 3.11/Linux, CUDA 12.8 locks (`requirements-v15-cuda.lock`, `-v15.lock`, `-head-v15.lock`), pinned official Transformers fork, `swiss-ai/Apertus-v1.5-8B` (revision `a411d838…`, ≈18.4 GB, six SHA-verified shards; gated: needs approved Hugging Face access and an `HF_TOKEN` for the first download only), E5-small (revision `614241f6…`), then `serve_v15.py --method head --head track_2a/deployment/head-v15.json`. We measured on A40 / RTX A6000 GPUs with 48 GB VRAM; this is our hardware, not an organizer requirement.
- **Audit trail:** model, data and PDF manifests, input contracts, registered protocols, head SHA-256 and lease costs are archived under `track_2a/experiments/`.

Data, models, license

OSTswiss/MNLIoverSwissVotingBooklets (1,488 examples, DE/FR/IT, all nine document–claim language pairs; split 902 train / 276 validation / 310 internal holdout by voting date and connected claims). Apertus v1.5 (Apache-2.0 with access/AUP conditions), multilingual-e5-small (MIT), pypdf (BSD-3). Booklet content © Federal Chancellery. Code Apache-2.0; this report CC-BY-4.0.

Repository: https://github.com/hallzyx/apertus · full technical report: `track_2a/technical_report.md` · narrated demo: `docs/demo/`